Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

MAKEUP END SEMESTER EXAMINATIONS, DECEMBER 2024

MODERN POWER CONVERTERS [ELE 4085]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: DECEMBER 2024** **Max. Marks: 50**

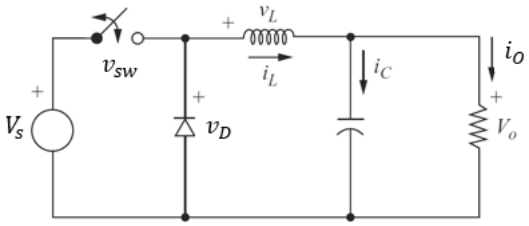
Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. A flyback converter is working in discontinuous current mode with source voltage 24 V, duty cycle 50%, load resistance of 10 Ω, switching frequency of 20 kHz and magnetizing inductor L_m of the transformer model 500uH. Determine the output voltage. Analyze the current waveform of L_m and source. **(03)**

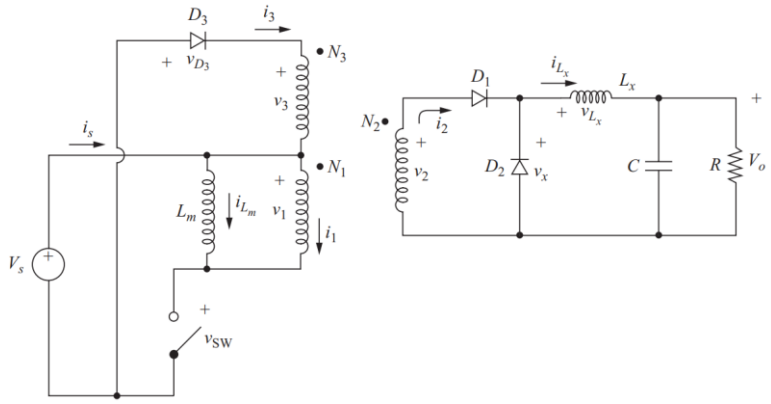
1B. Make a technical comparision between AT and ATX power supply **(03)**

1C Analyze the waveforms of inductor current, inductor voltage and capacitor current of the DC-DC converter given below. Derive an expression to estimate the output voltage.



(04)

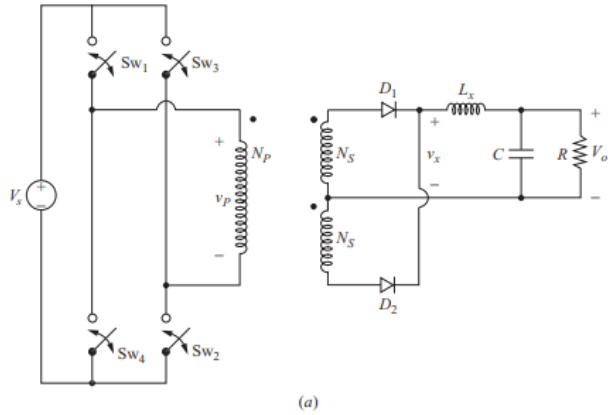
2A. Explain and analyze the need for tertiary winding in the DC-DC converter. Derive its output voltage expression.



(04)

2B. Make a technical comparison with respect to the filter size, voltage stress on switch and switching control of the push pull converter and forward converter with appropriate justifications. **(03)**

2C. Discuss the working of a DC-DC converter shown below and examine the voltage waveform v_x. **(03)**



(a)

3A. Classify the different soft switching techniques that have been developed to enhance the efficiency of the converters. **(02)**

3B. With the neat circuit diagram and waveform of resonant inductor current and capacitor voltage, explain the working of series loaded resonant (SLR) converter with leading power factor. **(04)**

3C. A parallel resonant half bridge DC-DC converter has the following parameters: input voltage: 100V, load resistance 10Ω. The switching frequency is 20% excess of the resonant frequency of 100 kHz. Determine the resonant tank component values Lr and Cr if its impedance is 5 Ω and output voltage. **(04)**

4A. Develop a small signal ac model of the buck-boost converter with step-by-step procedure. Assume ideal components and continuous conduction mode. **(04)**

4B. With a neat diagram, explain a negative feedback control switching regulator system for DC-DC buck converter. **(03)**

4C. With circuit diagram and necessary waveforms explain and analyze the working of Zero Voltage Switching (ZVS) resonant converter. **(03)**

5A. Discuss the mechanical parameters considered in inductor design. **(03)**

5B. Determine the inductance of the metal-core coil with length 1m and radius 2mm, number of turns be 200 and relative permeability 2000. **(02)**

5C. Design an inductor for the buck converter with following specifications: Switching frequency is 40 kHz, Output voltage = 10V, Input voltage = 15 V ± 10 %, Output Current, Io = 5A, Current ripple is 10% of Io, Bm = 0.2 T for ferrite, J = 5A/mm^2, Kc = 1.00

Core	Mean length (mm)	Core cross-section (Ac*100mm^2)	Window area (Aw*100 mm2)	Area Product (Ap*10^ 4 mm^2)	AL
EE20/10/5	38	0.31	0.478	0.149	1624
EE25/9/6	51.2	0.40	0.78	0.312	1895
EE25/13/7	52	0.55	0.87	0.478	2285
EE30/15/7	56	0.597	1.19	0.71	3025
EE36/18/11	70.6	1.31	1.41	1.847	4200
EE42/21/9	77.6	1.07	2.56	2.739	2613
EE42/21/15	93	1.82	2.56	4.659	4778
EE42/21/20	99	2.35	2.56	6.016	6231
EE65/32/13	150	2.66	5.37	14.284	4833

(05)